Confidence-Bounded Physical Feasibility Ranking for Neutronic Screening of QUBO-Feasible Fuel Layouts Using OpenMC

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***Abstract*—Screening pipelines for discrete engineering designs must distinguish encoded feasibility from declared physical limits while preserving unresolved ties and rejecting incomplete evidence. We formulate Confidence-Bounded Physical Feasibility Ranking (CB-PFR) as an auditable decision-support layer that computes per-metric operational bounds, normalized violations, and feasibility-first ranks with explicit tie groups. Four formal properties are stated and tested: violations are monotone in a common uncertainty multiplier; ranks and tie groups are invariant to input permutation; missing or non-finite required values fail closed; and setting the multiplier to zero recovers the corresponding point-estimate feasibility-first rule. We apply the workflow retrospectively to six enumerated two-dimensional multigroup C5G7-derived layouts (18 OpenMC runs and four reflection classes) and audit two three-seed UMMU protocols. At the illustrative threshold $F_{\max} = 1.35$, two classes pass and two fail; all k_{eff} bounds pass, so the decision in this case is driven by the relative fission-rate peaking proxy. Welch intervals for mirror-label differences are wide and do not establish statistical equivalence. In the 100-inactive-batch UMMU runs, between-run SD is 0.77 pcm against approximately 9.45 pcm pooled within-run SD, and repeated entropy-mesh warnings remain unresolved. The evidence supports a formal, auditable decision-support workflow, not calibrated confidence coverage, benchmark validation, physical optimality, comparative superiority, or quantum advantage.**

***Index Terms*—CB-PFR, decision support, C5G7 benchmark, OpenMC, QUBO, reproducible research software**

I. Introduction

A decision-support pipeline must expose why a candidate passed, failed, or remained tied rather than hide those choices in an objective value. Constrained optimization commonly uses feasibility-first rules: feasible solutions dominate infeasible ones, and infeasible solutions are compared by a declared violation measure [1]. Stochastic ranking offers a different way to balance objective and constraint information [2]. Reactor loading-pattern studies have also used genetic algorithms, simulated annealing, hybrid schemes, and problem-specific encodings [3]–[7], while recent work combines adjoint-based physical information with evolutionary search [8]. These works provide methodological context, not performance baselines for this small case.

This paper frames CB-PFR as a transparent decision-support component after candidate generation and physical evaluation, not as a new general feasibility-first principle. Its bounded contribution is an explicit operational rule for uncertainty-expanded constraints, noncompensatory feasibility, stable rank/tie-group output, and fail-closed handling of absent evidence. The study makes three testable claims: larger uncertainty multipliers cannot reduce declared violations; input permutation cannot change substantive ranks or tie groups; and invalid required observations cannot be selected as feasible. Four software properties and corresponding unit tests are provided. The case study enumerates all six material assignments; no pre-physics optimizer or annealer was run, and no superiority claim is made.

II. Case Study and Methods

A. Candidate Space and Composition QUBO

The two-dimensional model fixes five reflector positions and varies four fuel positions $p_{00}, p_{01}, p_{10}, p_{11}$. Each fuel position receives either MOX or UO_2, with exactly two of each. The six label strings are MMUU, MUMU, MUUM, UMMU, UMUM, and UUMM. Let $y_i \in \{0, 1\}$ indicate MOX at position i. The composition-only penalty is

$$E(\mathbf{y}) = A\left(\sum_{i=1}^{4} y_i - 2\right)^2, \qquad A > 0. \tag{1}$$

Using $y_i^2 = y_i$, its expansion is

$$E(\mathbf{y}) = A\left(4 - 3\sum_i y_i + 2\sum_{i<j} y_i y_j\right). \tag{2}$$

Every feasible assignment has $\sum_i y_i = 2$, hence $E = 0$. This algebraic degeneracy is a limitation of the stated composition-only objective, not of QUBO in general. The study is a complete enumeration of a small design space and does not test whether a pre-physics optimizer is useful at larger scales.

B. Symmetry and Neutronic Metrics

The declared 3×3 position map places the four variable fuel locations at the upper-left 2×2 positions and the five reflectors elsewhere. Reflection across the main diagonal swaps p_{01} and p_{10}. Thus MMUU/MUMU form class S1, MUUM is S2, UMMU is S3, and UMUM/UUMM form S4. This class reduction depends on the documented mapping and symmetric model assumptions.

OpenMC is used for multigroup neutron-transport and tally calculations [9]; the C5G7 MOX specification provides the benchmark context [10]. For the four variable fuel regions, let R_i be the assembly-integrated fission tally. The relative peaking proxy is

$$F = \frac{\max(R_1, R_2, R_3, R_4)}{\frac{1}{4}\sum_{i=1}^{4} R_i}. \tag{3}$$

F is dimensionless and relative. It is not absolute power, pin power, a thermal-hydraulic quantity, or an approved safety criterion. Source-distribution entropy and k-generation traces are used as descriptive diagnostics; monitoring source convergence is important in eigenvalue calculations, but no single scalar diagnostic alone certifies convergence [11].

C. CB-PFR Rule

For candidate or symmetry class i and metric j, let $\bar{x}_{ij}$ be the observed estimate and $s_{ij} \geq 0$ the supplied uncertainty summary. Under a common nonnegative multiplier z, CB-PFR forms operational bounds

$$B_{ij}^{-}(z) = \bar{x}_{ij} - z s_{ij}, \quad B_{ij}^{+}(z) = \bar{x}_{ij} + z s_{ij}. \tag{4}$$

For declared lower and upper limits L_j, U_j, the violation is

$$v_{ij}(z) = \max(0, L_j - B_{ij}^{-}(z)) + \max(0, B_{ij}^{+}(z) - U_j), \tag{5}$$

including only terms for limits that are declared. The normalized aggregate is $V_i(z) = \sum_j w_j v_{ij}(z)/d_j$, with nonnegative weights w_j and positive scales d_j. These operational bounds are not calibrated confidence intervals.

Physical ranking uses a feasibility-first score: QUBO-feasible candidates precede QUBO-infeasible candidates; then physically feasible candidates precede physically infeasible candidates; among infeasible candidates the smaller declared V_i is preferred; among feasible candidates the QUBO energy remains the declared objective. The objective is not used to break ties among candidates with the same infeasible violation. Candidates with equal substantive scores receive the same rank and tie-group. Candidate ID is used only for deterministic display within a tie, never as a preference. Missing or invalid required observations create infinite violation and a failing status.

D. Formal Properties

Proposition 1 (Monotone uncertainty penalty). For fixed estimates, nonnegative uncertainties, limits, weights, and positive scales, $V_i(z)$ is nondecreasing in $z \geq 0$. *Proof.* Each term is the maximum of zero and an affine function with nonnegative slope in z; sums with nonnegative coefficients preserve monotonicity. Therefore the set of candidates satisfying every declared limit cannot expand as z increases.

Proposition 2 (Permutation invariance and tie preservation). Reordering candidate inputs leaves each candidate's substantive score, rank, and tie-group unchanged. *Proof.* Scores depend only on candidate data and declared limits. Sorting uses that score and then candidate ID only for display; ties are formed from the score before this display ordering, so an input index cannot split a tie.

Proposition 3 (Fail-closed inputs). A missing or non-finite required estimate cannot be marked feasible or selected as conservatively feasible. *Proof.* Invalid observations receive infinite normalized violation and a failing metric status; the physical feasibility indicator is one, including when a metric weight is zero.

Proposition 4 (Point-estimate limit). Setting $z = 0$ gives the same bounds and substantive ordering as a point-estimate feasibility-first rule using the same limits, aggregate-violation definition, and QUBO objective. *Proof.* Both bounds collapse to $\bar{x}_{ij}$, so each metric status and violation is identical. The score tuple is consequently identical. This is a Deb-style rule with the declared normalized aggregate violation; it is not a claim that all Deb implementations use the same aggregation.

The anonymous supplement supplies unit tests for these four properties. They test software semantics, not the physical validity of a source model or the calibration of uncertainty summaries.

E. Archived Campaigns and Exploratory Limits

The six-layout campaign contains 18 recorded multigroup OpenMC runs: six labels across seed rounds 820000, 830000, and 840000. Each used the documented seven-group C5G7 energy structure, 7,000 total batches, 500 inactive batches, 10,000 particles per batch, and two threads. The records contain unique seeds and statepoint hashes and a common multigroup cross-section (MGXS) hash. These archive checks support traceability, but the model geometry/material mapping and spatial-tally correspondence have not been independently reviewed against the primary specification.

A separate three-seed UMMU repeatability assessment used the same total batch count (7,000) and particles per batch (10,000) under two protocols: 500 inactive batches in one archived run set and 100 inactive batches in a later run set. The 500-inactive simulations completed and their raw HDF5 audits passed, but the initial aggregation job stopped because the source-convergence report for seed 720000 was absent. The archived statepoints were subsequently inspected directly for active-batch entropy and k-generation traces. The later 100-inactive run set produced these reports but logged repeated warnings that fission-source sites fell outside the entropy mesh. The two protocols are therefore reported separately rather than pooled or presented as interchangeable replications.

No design-specific acceptance rule was independently approved before the six-layout campaign. Therefore, $k_{\text{eff}} \in [1.14, 1.21]$, $F_{\max} = 1.35$, $z = 1$, scales $d_k = 0.01$, $d_F = 0.1$, and unit weights are explicitly retrospective illustrative settings. They exercise the software pathway only. A sweep over $F_{\max} \in \{1.30, 1.32, 1.35, 1.80, 1.82, 1.87, 1.90\}$ and $z \in \{0, 1, 2\}$ is sensitivity analysis, not confirmation.

III. Results

A. Symmetry-Class Aggregation

Table I summarizes class-level run results. $s_{k,\mathrm{run}}$ is the sample standard deviation of run-level k_{eff} means within each class; $\bar{s}_{k,\mathrm{within}}$ is the mean reported within-run standard deviation. For the illustrative CB-PFR screen, $s_k^* = \max(s_{k,\mathrm{run}}, \bar{s}_{k,\mathrm{within}})$. The s_F column is the sample standard deviation of run-level F observations pooled within the class. Values are descriptive, not estimates of calibrated confidence coverage.

The measured F means separate the four classes descriptively, while mirror members within S1 and S4 should not be assigned distinct physical ranks from their small sampling differences. Across the 18 layout runs, first-to-last active-quartile mean shifts ranged from -1.81×10^{-3} to $+3.11 \times 10^{-3}$ for source entropy and from -1.07×10^{-3} to $+1.16 \times 10^{-3}$ for k-generation. These summaries are diagnostic only: no convergence tolerance was set prospectively, and small quartile shifts do not certify source convergence.

Reflection defines the theoretical classes, but this does not itself establish empirical equivalence between independently simulated mirror labels. Table II reports Welch 95% intervals for the run-level differences between the two mirror pairs. With only three runs per label, the intervals are wide. Because no equivalence margin was specified before data acquisition, these are descriptive intervals rather than a confirmatory equivalence test; the manuscript makes no statistical-equivalence claim. Across the 18 layout runs, first-to-last active-quartile mean shifts ranged from -1.81×10^{-3} to $+3.11 \times 10^{-3}$ for source entropy and from -1.07×10^{-3} to $+1.16 \times 10^{-3}$ for k-generation. These summaries are diagnostic only: no convergence tolerance was set prospectively, and small quartile shifts do not certify source convergence.

B. CB-PFR Status and Threshold Sensitivity

For $z = 1$, all k_{eff} bounds fall within $[1.14, 1.21]$, so $V_k = 0$ for every class. The k_{eff} constraint is therefore inactive in this case: it contributes no information to the pass/fail split. Table III exposes every step: bounds, per-metric violations, total violation, status, and the rank/tie-group resulting from the declared feasibility-first rule. At the retrospective $F_{\max} = 1.35$ threshold, S1 and S2 pass and remain tied because both have zero violation and the same QUBO energy; S4 has the smaller violation among failures and ranks above S3. These are software-pathway results, not engineering decisions.

Table IV reports counts as *label strings / symmetry classes*; for example, S1 contributes two label strings but one physical class. The $z = 0$ and $z = 1$ counts coincide across this tested grid; $z = 2$ changes the count only at $F_{\max} = 1.87$. The passing count is therefore dependent on the retrospective cutoff, which is not independently justified as a physical limit.

C. UMMU Repeatability and Protocol Audit

Table V compares two archived three-seed UMMU run sets. Both used 7,000 total batches, 10,000 particles per batch, the same multigroup cross-section hash, unique random seeds, and passing raw HDF5/group-structure audits; the inactive-batch setting differs. The reference $k_{\mathrm{ref}} = 1.18655$ and its standard uncertainty $u_{\mathrm{ref}} = 9.5 \times 10^{-5}$ (9.5 pcm) are the published C5G7 Monte Carlo reference values reported for the benchmark [10], [12]; they were not recomputed as an independent reference in this project. The differences are expressed in absolute pcm as $(\bar{k} - k_{\mathrm{ref}}) \times 10^5$. This reference uncertainty is not a tolerance for accepting the present layout model.

For a descriptive combined-uncertainty score only, we use

$$u_{\mathrm{run}} = \frac{\max(s_{\mathrm{between}}, s_{\mathrm{within}})}{\sqrt{n}}, \qquad z_{\mathrm{comb}} = \frac{\bar{k} - k_{\mathrm{ref}}}{\sqrt{u_{\mathrm{ref}}^2 + u_{\mathrm{run}}^2}},$$

where s_{between} is the sample SD of run means and s_{within} is the pooled within-run reported SD. This conservative operational convention yields $z_{\mathrm{comb}} = -0.434$ for the 500-inactive set and -0.698 for the 100-inactive set. It is a descriptive project calculation, not an acceptance criterion or evidence of benchmark validation.

The two protocol summaries must not be merged: between-run SD is about 1.63 times the pooled within-run SD for Protocol A but only 0.081 times that SD for Protocol B. The latter is a pronounced under-dispersion relative to the run-level uncertainty estimates and requires investigation rather than interpretation as unusually strong repeatability. Active-trace first-to-last-quartile k-generation shifts are $+24.0$ to $+65.1$ pcm for Protocol A and -16.1 to $+23.5$ pcm for Protocol B. The corresponding entropy shifts are -0.00195 to -0.00007 and -1.456 to -0.475, respectively; the 100-inactive logs also recorded repeated source sites outside the entropy mesh. These traces do not certify source convergence. Neither protocol is treated as a validated canonical estimate; model/tally review and convergence investigation remain necessary.

IV. Discussion

The results illustrate three issues relevant to a decision-support reviewer. First, the composition-only QUBO enforces material count but ties all six feasible labels by construction. Second, the rank/tie-group output makes the decision rule inspectable: at $z = 1$, S1 and S2 tie as passing classes; S4 is the less-violating failed class; S3 is last. Third, mirror symmetry is a property of the idealized model mapping, not proof that the two independently simulated labels are statistically equivalent. The Welch intervals are too wide to establish tight equivalence, and no prospectively declared margin is available. The sensitivity plot also shows uncertainty expansion changes a decision only in a narrow threshold region around S3; with $F_{\max} = 1.35$, the current decision is unchanged across $z = 0, 1, 2$.

The paper does not claim that the selected proxy is an engineering acceptance criterion. A stronger confirmatory study would require physical limits justified independently of this campaign, a prospectively specified analysis, independent model/tally review, and an adequate comparison against simple methods. A larger candidate space and a meaningful physics-sensitive pre-physics objective would be required

TABLE I
SIX-LAYOUT MULTIGROUP OPENMC CAMPAIGN AGGREGATED BY DIAGONAL-REFLECTION CLASS.

Class	Layout labels	n	Mean k_{eff}	$s_{k,\text{run}}$	$\bar{s}_{k,\text{within}}$	Mean F	s_F
S1	MMUU / MUMU	6	1.143686	7.10×10^{-5}	9.54×10^{-5}	1.295867	0.000960
S2	MUUM	3	1.156263	8.59×10^{-5}	9.53×10^{-5}	1.316149	0.001297
S3	UMMU	3	1.186499	1.40×10^{-4}	9.70×10^{-5}	1.867100	0.001541
S4	UMUM / UUMM	6	1.204665	7.94×10^{-5}	9.71×10^{-5}	1.816406	0.000673

TABLE II
EXPLORATORY INDEPENDENT-RUN MIRROR-LABEL DIFFERENCES (FIRST LABEL MINUS SECOND). WELCH 95% INTERVALS USE THREE RUN MEANS PER LABEL.

Pair	Metric	Mean difference	95% Welch interval
MMUU − MUMU	k_{eff} (pcm)	-2.81	$[-22.88,\ 17.26]$
MMUU − MUMU	F	-9.95×10^{-4}	$[-3.15,\ 1.16]\times10^{-3}$
UMUM − UUMM	k_{eff} (pcm)	$+4.67$	$[-16.52,\ 25.86]$
UMUM − UUMM	F	$+6.28\times10^{-4}$	$[-1.47,\ 2.73]\times10^{-3}$

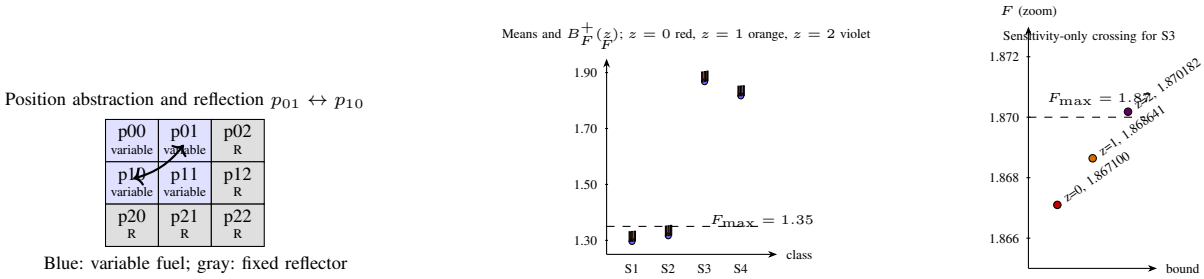


Fig. 1. (Left) Four variable fuel positions and five fixed reflector positions; diagonal reflection exchanges p_{01} and p_{10}. (Right, first panel) Class means and upper operational bounds $B_F^+(z)$ for $z = 0, 1, 2$ on the full range; uncertainty bounds are visually small on this scale. (Right, second panel) A zoomed, sensitivity-only view shows that the S3 decision changes at $F_{\max} = 1.87$ between $z = 1$ and $z = 2$. Neither threshold is an engineering limit.

TABLE III
ILLUSTRATIVE CB-PFR DECISION TRACE: $k \in [1.14, 1.21]$, $F_{\max} = 1.35$, $z = 1$. VIOLATIONS ARE NORMALIZED BY $d_k = 0.01$ AND $d_F = 0.1$.

Class	B_k^-	B_k^+	B_F^+	V_k	V_F	V_{total}	Rank / tie group	Status
S1	1.143591	1.143781	1.296827	0.0000	0.0000	0.0000	1 / G1	Pass
S2	1.156168	1.156358	1.317446	0.0000	0.0000	0.0000	1 / G1	Pass
S3	1.186359	1.186639	1.868641	0.0000	5.1864	5.1864	4 / G3	Fail
S4	1.204568	1.204762	1.817079	0.0000	4.6708	4.6708	3 / G2	Fail

TABLE IV
RETROSPECTIVE SENSITIVITY SWEEP: PASSING LABEL STRINGS / SYMMETRY CLASSES.

$F_{\max}$	$z = 0$	$z = 1$	$z = 2$
1.30	2 / 1	2 / 1	2 / 1
1.32	3 / 2	3 / 2	3 / 2
1.35	3 / 2	3 / 2	3 / 2
1.80	3 / 2	3 / 2	3 / 2
1.82	5 / 3	5 / 3	5 / 3
1.87	6 / 4	6 / 4	5 / 3
1.90	6 / 4	6 / 4	6 / 4

to evaluate claims about optimizer utility. The rule is positioned alongside, rather than claimed superior to, established constraint-handling, stochastic-ranking, and simulation ranking-and-selection methods [1], [2], [13]. For uncertainty terminology, within-run Monte Carlo standard uncertainties are distinguished from observed between-run dispersion, consistent with measurement-uncertainty guidance [14]. The reference value $k_{\text{ref}} = 1.18655$ is the published C5G7 Monte Carlo reference, not a validation result for the present layout model [10], [12]. The contribution is the tested software decision trace and evidence-bounded audit, not priority for feasibility-first ranking.

V. CONCLUSION

This paper reports an exploratory, evidence-audited CB-PFR workflow on six enumerated C5G7-derived fuel layouts. The composition QUBO assigns zero energy to each feasible layout; diagonal-reflection symmetry reduces six labels to four classes. Under a retrospective illustrative screen, two classes pass and two fail, while the sensitivity table makes

TABLE V
SEPARATE UMMU RUN PROTOCOLS AND REPEATABILITY SUMMARIES.

Protocol	Inactive batches	Mean k_{eff}	Δk (pcm)	Between-run SD (pcm)	Pooled within-run SD (pcm)
A	500	1.18649321	-5.68	15.57	9.58
B	100	1.18647358	-7.64	0.77	9.45

threshold dependence explicit. The separate UMMU run sets give absolute reference differences of -5.68 pcm and -7.64 pcm under 500- and 100-inactive-batch protocols, respectively. Their repeatability diagnostics differ materially, and neither establishes benchmark acceptance. The evidence supports workflow traceability and transparent handling of ties, not physical optimality, calibrated confidence coverage, full validation, or method superiority.

SOFTWARE AND DATA AVAILABILITY

The accompanying anonymous-review supplement contains the 18-run six-layout table, six UMMU run-level records with statepoint hashes, source-trace diagnostics, a reproducible mirror-label comparison script, and unit tests for the formal ranking properties. The unit tests verify decision-rule behavior; they do not certify source convergence or physical validity. The project identity and permanent public release are withheld from this double-blind manuscript; no DOI or public release is claimed in this review version.

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